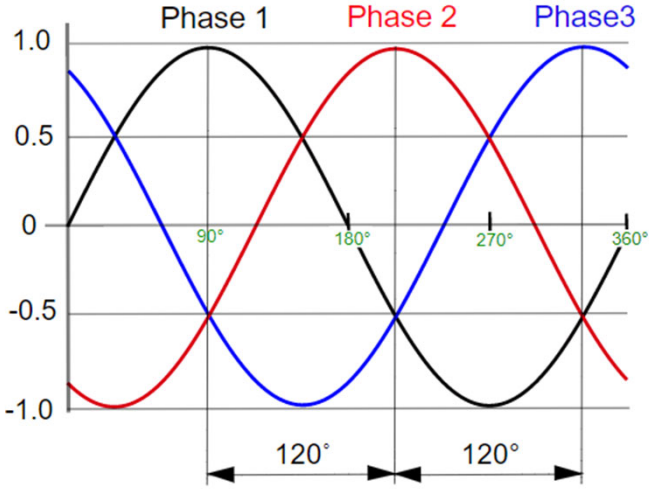
ECE 330

Chapter 2

Phasors and three-phase circuits



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Learning Objectives: you will be able to

- Convert time-domain sinusoidal sources to phasor form using **RMS values**
- Use **RMS phasors** to analyze steady-state AC circuits
- Define and calculate **real (P), reactive (Q)**, **apparent (S)** and **complex power ($\bar{S}$)**
- Define **power factor (PF),** and classify loads as having **leading, lagging, or unity** power factor.
- Use the **power triangle** to compute power and power factor.
- Explain the motivation for **power factor correction** in AC power systems, and calculate the required reactive compensation to achieve a desired PF.
- Explain the motivation for using three-phase power systems.
- Identify **Wye (Y)** and **Delta (Δ)** three-phase connections and analyze balanced three-phase loads connected in Y or Δ configurations.
- Develop **per-phase equivalent circuits** for balanced Y- and Δ-connected loads and use single-phase circuit analysis techniques to solve three-phase power problems

Review of Phasors

- ECE 210

$$A(t) = A_m \cos(\omega t + \theta)$$

magnitude; phase shift

⇓ phasors

$$A = A_m \angle \theta$$

- ECE 330

$$A(t) = A_m \cos(\omega t + \theta)$$

⇓ phasors

$$\bar{A} = \frac{A_m}{\sqrt{2}} \angle \theta$$

A_{rms}

Converting from sine to cosine

$\sin x = \cos(x - 90^\circ)$

$$v(t) = \sqrt{2}(10)\sin\left(wt + 30^\circ\right) =$$

$= \sqrt{2}(10)\cos(wt + 30^\circ - 90^\circ)$

-60°

⇓ phasors

$\bar{V} = \frac{\sqrt{2}(10)}{\sqrt{2}} \angle -60^\circ$ Volts

fix it by $\pm 180^\circ$ shift

$$i(t) = -\sqrt{2}(5)\sin\left(wt - 20^\circ\right) =$$

$= \sqrt{2}(5)\cos(wt - 20^\circ - 90^\circ + 180^\circ)$

70°

⇓ phasors

$\bar{I} = 5 \angle 70^\circ$ Amps

Example 1: write as phasors

$$i(t) = \sqrt{2}(5)\cos(wt - 20^\circ)$$

A) $\bar{I} = \sqrt{2}(5)\angle -20^\circ$

B) $\bar{I} = (5)\angle -110^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle -110^\circ$

D) $\bar{I} = (5)\angle -20^\circ$

Example 2: write as phasors

$$i(t) = -\sqrt{2}(5)\sin\left(wt + 30^\circ\right) = 5\cos\left(wt + 30^\circ - 90^\circ + 180^\circ\right)$$

$\sqrt{2}$ — 120°

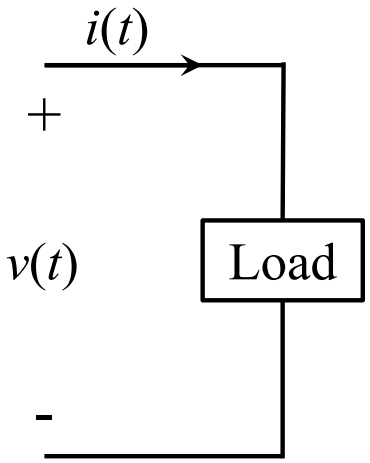
A) $\bar{I} = (5)\angle 120^\circ$

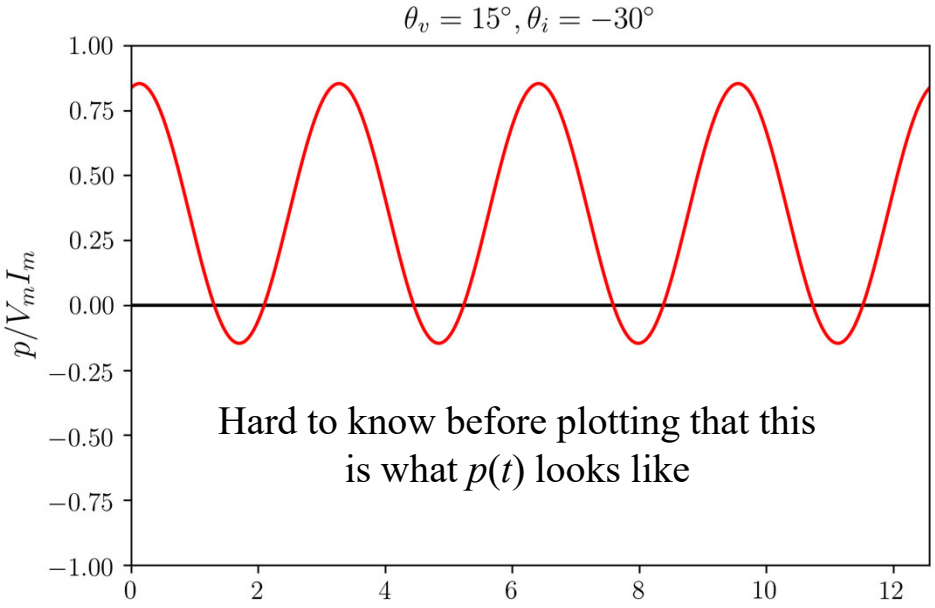
B) $\bar{I} = \sqrt{2}(5)\angle 30^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle 120^\circ$

D) $\bar{I} = (5)\angle 30^\circ$

Single Phase: 1 voltage and 1 current





- $p(t) = v(t)\,i(t)$
- $v(t) = V_m \cos(\omega t + \theta_v)$

  $i(t) = I_m \cos(\omega t + \theta_i)$
- $p(t) = V_m I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i)$

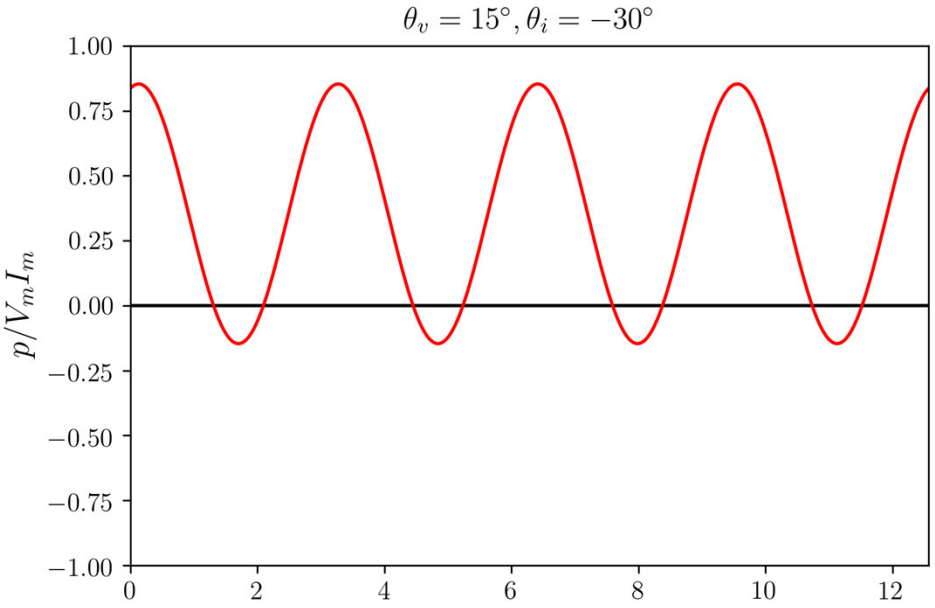
Simplifying p(t)

$$\cos(A)\cos(B) = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$$

$$p(t) = \frac{V_m I_m}{2}\left(\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)\right)$$

$\frac{V_m I_m}{2} = V_{rms} I_{rms}$

- $V_{rms} = \frac{V_m}{\sqrt{2}}$
- $I_{rms} = \frac{I_m}{\sqrt{2}}$



Average Power: P

$$P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{T}\int_0^T V_{rms} I_{rms}\left(\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)\right)dt =$$

$$= \frac{1}{T}\cdot T\cos(\theta_v - \theta_i)\, V_{rms} I_{rms}$$

$$\boxed{P = V_{rms} I_{rms}\cos(\theta_v - \theta_i)}$$

$[W]$

- average (real, active) power "doing the work"

Reactive (Q) and Complex ($\bar{S}$) power

- $P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$ – real power [W]
- $Q = V_{rms} I_{rms} \sin(\theta_v - \theta_i)$ – reactive power [VAr]

voltamperes reactive

$\Downarrow$

- $P = \mathrm{Re}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$
- $Q = \mathrm{Im}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$

$\Downarrow$

- $P = \mathrm{Re}\{\underbrace{V_{rms} e^{j\theta_v}}_{\bar{V}} \cdot \underbrace{I_{rms} e^{-j\theta_i}}_{\bar{I}^*}\} = \boxed{P = \mathrm{Re}\{\bar{V} \cdot \bar{I}^*\}}$
- $Q = \mathrm{Im}\{V_{rms} e^{j\theta_v} \cdot I_{rms} e^{-j\theta_i}\} = \boxed{Q = \mathrm{Im}\{\bar{V} \cdot \bar{I}^*\}}$

apparent power: $S = V \cdot I$

$S \angle \theta$

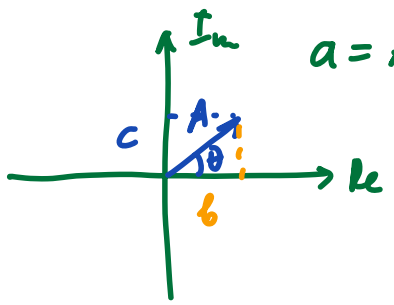
$\boxed{\bar{S} = \bar{V} \cdot \bar{I}^* = P + jQ}$ complex power [VA]

voltamperes

$(\theta_v - \theta_i)$: power factor angle

$\cos(\theta)$: power factor (PF) 0 – 1

Detour



$a = Ae^{j\theta}$

$b = \mathrm{Re}\{Ae^{j\theta}\} = A\cos\theta$

$c = \mathrm{Im}\{Ae^{j\theta}\} = A\sin\theta$

Example 3:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find P:

$P = \mathrm{Re}\{\bar{V}\cdot\bar{I}^*\} = \mathrm{Re}\{10\angle 15^\circ \cdot 5\angle 30^\circ\} = \mathrm{Re}\{50\angle 45^\circ\} = 35.35$ W

units, please ☺

Example 4:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find Q:

A) 50 W

B) 35.355 W

C) -12.94 VAR

D) 35.355 VAR

Example 5:

$v(t) = \sqrt{2}(10)\cos(\omega t + 15^\circ)$ ⇒ $\bar{V} = 10\angle 15^\circ$

$i(t) = \sqrt{2}(7)\sin(\omega t + 75^\circ)$ ⇒ $\bar{I} = 7\angle -15^\circ$

$= \sqrt{2}(7)\cos(\omega t + 75^\circ - 90^\circ)$

Find $\bar{S}$:

A) $70 + j0$ VA

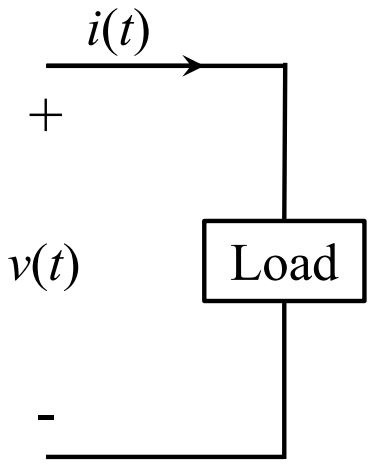
B) $35 - j60.62$ VA

C) $60.62 + j35$ VA

D) $121.24 + j70$ VA

① $\bar{S} = \bar{V} \cdot \bar{I}^* = 10\angle 15^\circ \cdot 7\angle 15^\circ = 70\angle 30^\circ$ VA

Another way to view complex power



$\bar{Z} = R + jX$ ← reactance

↑ resistance

① $\bar{S} = \bar{V} \cdot \bar{I}^* = \bar{Z}(\bar{I} \cdot \bar{I}^*) = \bar{Z} \cdot I^2$

↑ $\bar{V} = \bar{Z} \cdot \bar{I}$

② $\bar{S} = I^2(R + jX) = \underbrace{I^2 \cdot R}_{P} + j\underbrace{I^2 \cdot X}_{Q} = P + jQ$

P: power dissipated by resistors

Q: power generated or consumed by reactances

Note: we can show that $\theta = \theta_z$

$\bar{S} = \bar{Z} \cdot I^2 = I^2 z \angle\theta_z = S\angle\theta \Rightarrow \theta = \theta_z$

cap: $X_C = -\frac{1}{\omega C}$

$Q < 0$

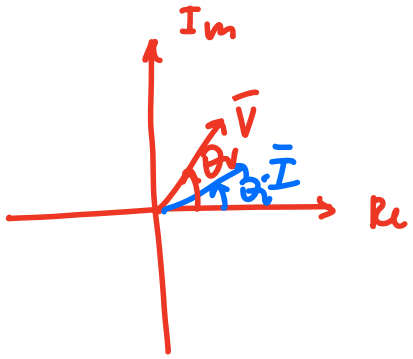
ind: $X_L = \omega L$

$Q > 0$

Recall: cosine is an even function

$$\cos(x)=\cos(-x)$$

PF = cos($\underbrace{\theta_v-\theta_i}_{\theta}$)



Current lags the voltage

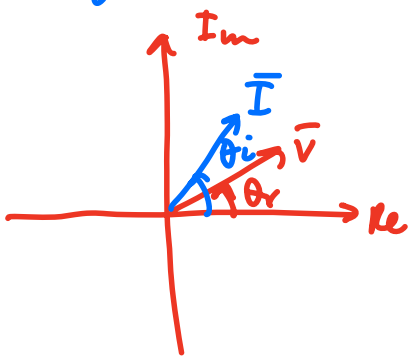
$\theta_v > \theta_i$

$\theta = \theta_v - \theta_i > 0$

Lagging PF

Note: voltage is a reference!

Note: PF is only defined for θ between -90° and 90°



current leads the voltage

$\theta_v < \theta_i$

$\theta < 0$

Leading PF

How do leading and lagging PF correspond to circuit?

Inductor

- $\bar{S}_L$ - ?
- $\bar{S}_L = \bar{V} \cdot \bar{I}^*$

$i(t) = I_m \cos(\omega t + \theta_i) \Rightarrow \bar{I} = I \angle \theta_i$

$v(t) = L \frac{di}{dt} \Rightarrow \bar{V} = L(j\omega \bar{I}) = \omega L I \angle \theta_i + 90^\circ$

$j = 1 \angle 90^\circ$

* $\frac{di}{dt} \rightarrow j\omega \bar{I}$

- $\bar{S}_L = \bar{V} \cdot \bar{I}^* = \omega L I^2 \angle \theta_i + 90^\circ - \theta_i$ (>0)

PF = $\cos(\theta) = 0$ lag

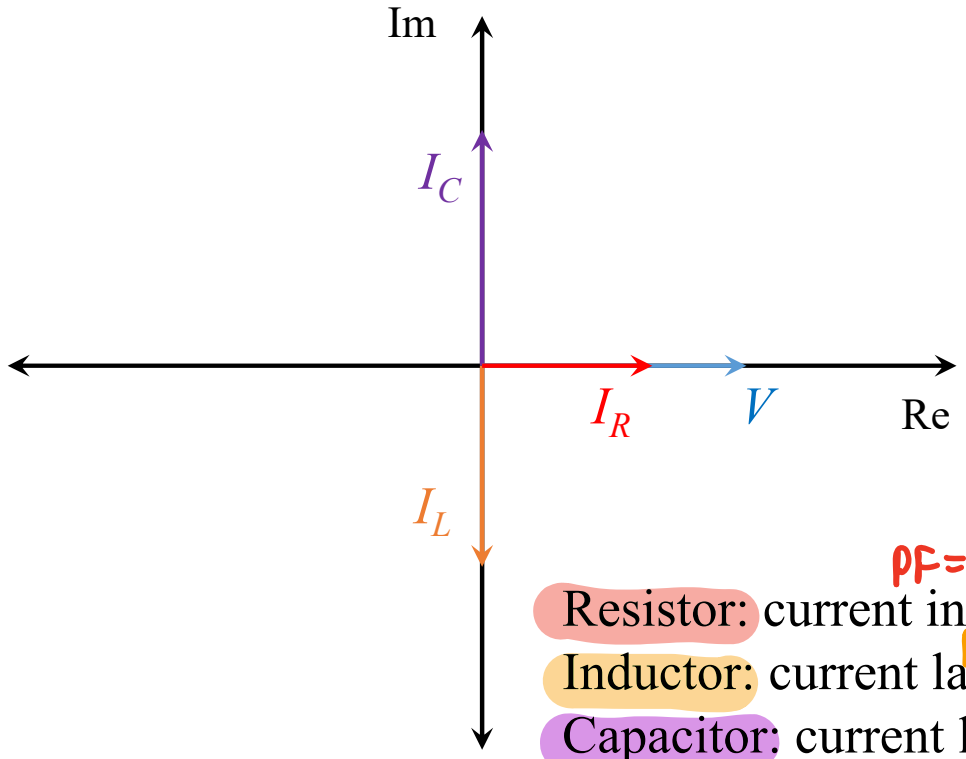
Capacitor

- $\bar{S}_C$ - ?
- $\bar{S}_C = \bar{V} \cdot \bar{I}^*$
- $v(t) = V_m \cos(\omega t + \theta_v) \Rightarrow \bar{V} = V \angle \theta_v$
- $i(t) = C \frac{dv}{dt} \Rightarrow \bar{I} = j\omega \bar{V} C$

- $\bar{S}_C = C\omega V^2 \angle \theta_v - \theta_v - 90^\circ$ (<0)

PF = 0 lead

Plotting in Complex Plane



PF = 1 (unity) Q = 0

Resistor: current in phase with voltage

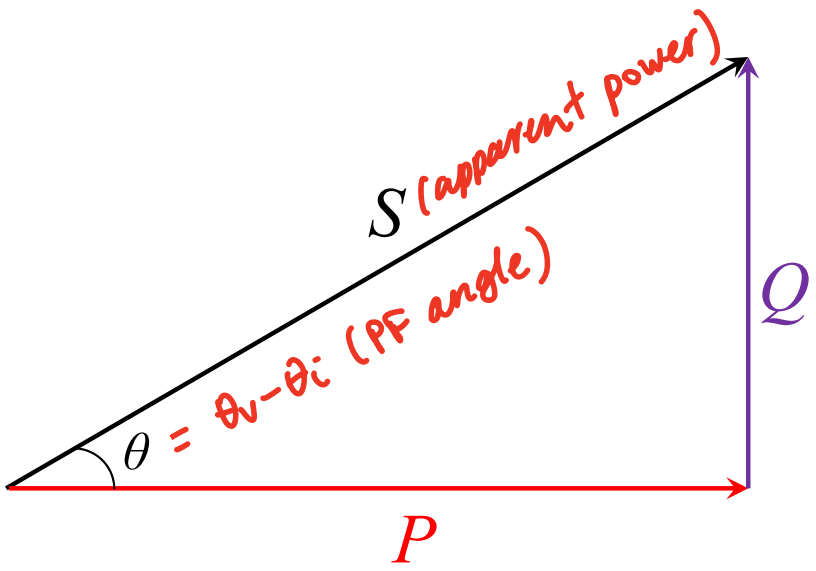
PF = 0 lag P = 0

Inductor: current lags voltage by 90°

Capacitor: current leads voltage by 90°

PF = 0 lead P −

Power Triangle: Lagging PF ($\theta > 0$) (inductive load)



- $\bar{S} = P + jQ = \bar{V} \cdot \bar{I}^* = S\angle\theta$

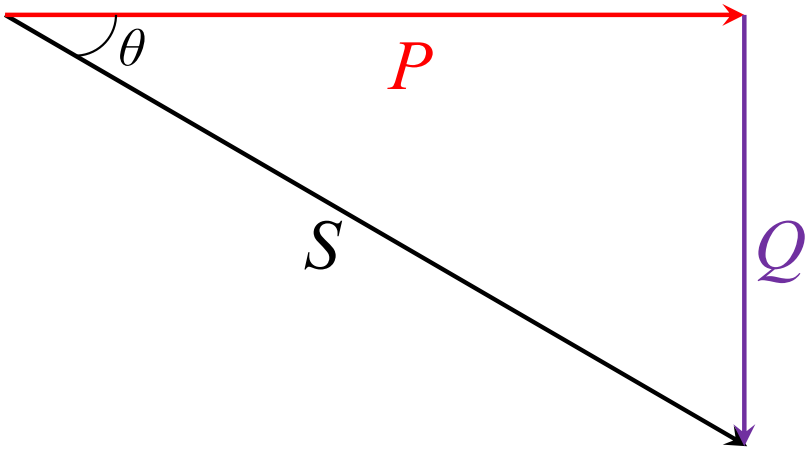
  [VA] [W] [VAR]

- $Q = VI\sin\theta$
- $P = S\cos\theta$

  $Q = S\sin\theta$

  $S = \sqrt{P^2 + Q^2}$

Power Triangle: Leading PF ($\theta < 0$) (capacitive load)



- $Q = VI\sin(\theta)$
- $P = S\cos\theta$

  $Q = S\sin\theta$

  $S = \sqrt{P^2+Q^2}$

Specifying Power

- 6 total quantities: V, I, PF, P, Q, S
- Need a reference: assume $\bar{V}$ is a reference $\Rightarrow \theta_V = 0$
- Use whatever is given to calculate unknown quantities

$\theta = \theta_V - \theta_i \Rightarrow \theta_i = -\theta$

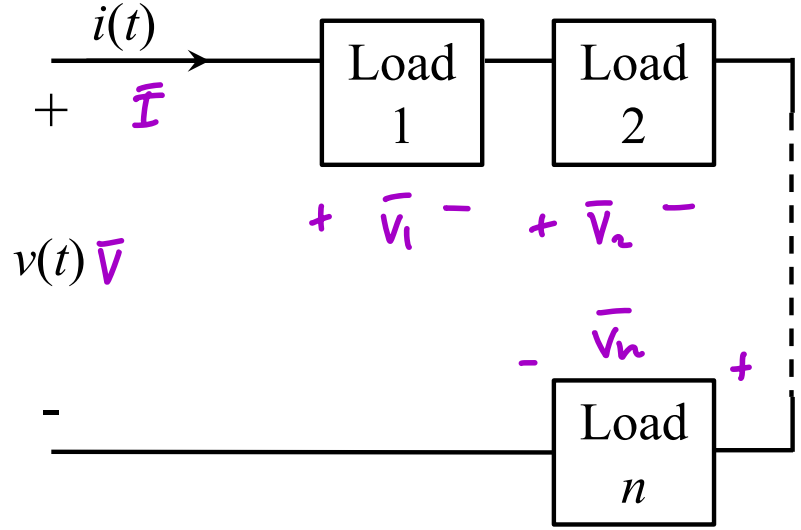
$\bar{I} = I\angle -\theta$

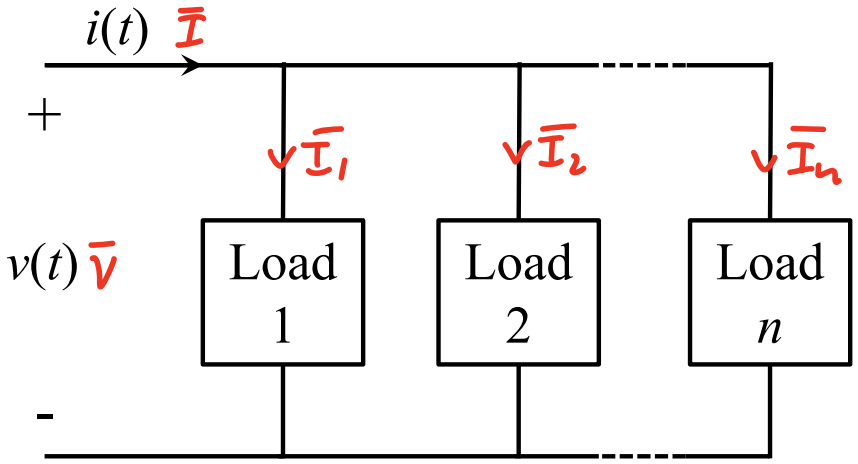
$$\theta = \begin{cases} \text{lag}: \theta > 0 \Rightarrow +\cos^{-1}(PF) \\ \text{lead}: \theta < 0 \Rightarrow -\cos^{-1}(PF) \end{cases}$$

- $P = S\cos\theta = S \cdot PF \Rightarrow S = \dfrac{P}{PF}$
- $Q = S\sin\theta = \dfrac{P}{PF} \cdot \sin\theta = P\tan\theta$

  ($PF = \cos\theta$)
- $S = \sqrt{P^2 + Q^2} = V \cdot I$

What if there are multiple loads?





- $\bar{S}_{tot} = \bar{V} \cdot \bar{I}^* = (\bar{V}_1 + \bar{V}_2 + .. + \bar{V}_n)\bar{I}^* =$

$= \bar{V}_1\bar{I}^* + \bar{V}_2\bar{I}^* + .. + \bar{V}_n \cdot \bar{I}^* = \bar{S}_1 + \bar{S}_2 + .. + \bar{S}_n$

($\bar{V}_1\bar{I}^* = \bar{S}_1$, $\bar{V}_2\bar{I}^* = \bar{S}_2$, $\bar{V}_n \cdot \bar{I}^* = \bar{S}_n$)

$$\bar{S} = \bar{S}_1 + \bar{S}_2 + .. + \bar{S}_n$$

$$P = P_1 + P_2 + .. + P_n$$

$$Q = Q_1 + Q_2 + .. + Q_n$$

- $\bar{S} = \bar{V} \cdot \bar{I}^*$

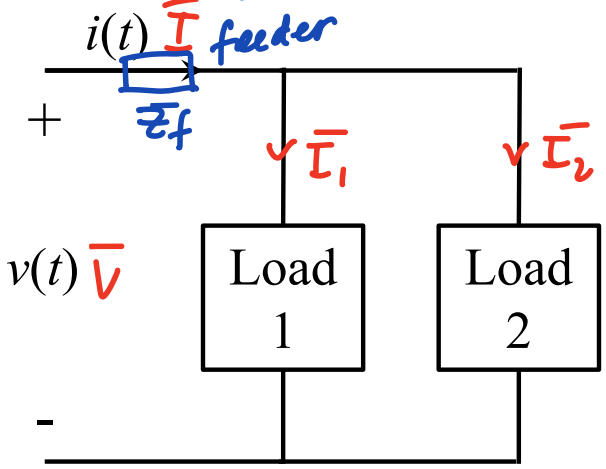
$\bar{I} = \bar{I}_1 + \bar{I}_2 + .. + \bar{I}_n$

- $\bar{S} = \bar{V}(\bar{I}_1 + \bar{I}_2 + .. + \bar{I}_n)^*$

$= \bar{V}\bar{I}_1^* + \bar{V}\bar{I}_2^* + ... + \bar{V}\bar{I}_n^*$

($\bar{V}\bar{I}_1^* = \bar{S}_1$, $\bar{V}\bar{I}_2^* = \bar{S}_2$, $\bar{V}\bar{I}_n^* = \bar{S}_n$)

Example 6: find S_{tot}



$$\bar{V} = 100\angle 10^\circ \text{ V}$$

$$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$$

$$\bar{I}_2 = 5\angle 40^\circ \text{ A}$$

$$\cdot S_{tot} = \bar{S}_1 + \bar{S}_2 = \bar{V}\cdot\bar{I}_1^* + \bar{V}\cdot\bar{I}_2^* =$$

$$= 100\angle 10 \cdot 10\angle 26.8^\circ + 100\angle 10 \cdot 5\angle -40^\circ =$$

$$= 1233 + j350 \text{ VA}$$

Purely reactive loads:

For ind and cap $P = 0$

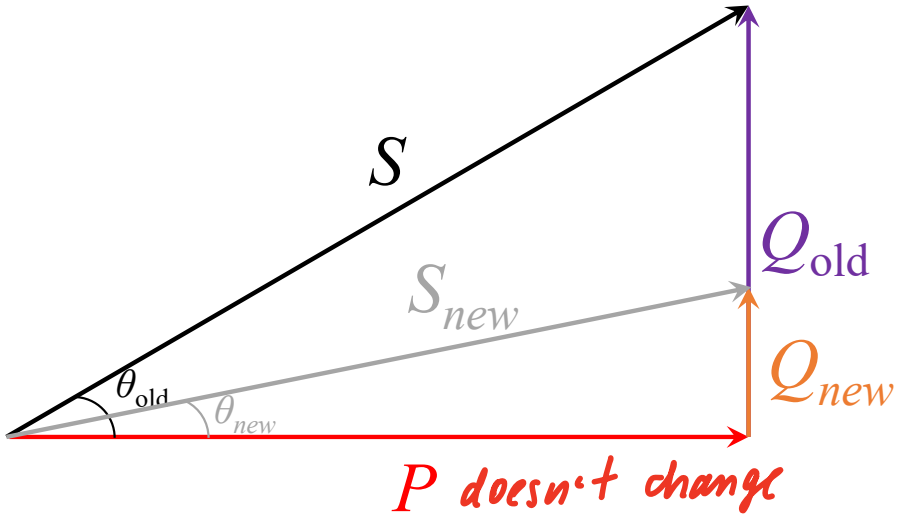
- Ind : $Q > 0$: consume reactive power
- Cap : $Q < 0$: produce reactive power

} PF correction : can supply Q locally by installing a capacitor in parallel with a load (ex. motor)

Why PF correction?

- $S = \sqrt{P^2 + Q^2} = V \cdot I$
- $I = \frac{S}{V}$ ⇒ if we keep V constant and S ↑ ⇒ I ↑ ⇒ greater losses ($I^2 R$) on the line. With PF correction, we minimize losses by lowering Q with Qc ⇒ I ↓.

Power Triangle: Power Factor Correction



- $Q_{old} + Q_c = Q_{new}$

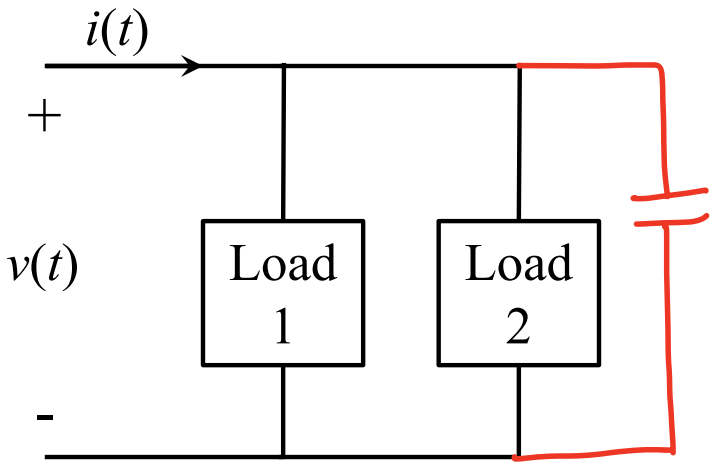
$Q_c = Q_{new} - Q_{old}$

↑ we know that $\frac{Q_{new}}{P} = \tan(\theta_{new})$ ← from PF new

P doesn't change

- Report as: add $|Q_c|$ VAR of capacitance

Example 7: PF correction



$\bar{V} = 100\angle 10^\circ \text{ V}$

$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$

$\bar{I}_2 = 5\angle 40^\circ \text{ A}$

From ex. 6

$\bar{S} = 1233 + j350 \text{ VA}$

P Qold

What should capacitor in parallel add to get PF = 0.98 lagging? (Q_C (VAR)-?)

① $Q_{old} + Q_C = Q_{new}$

$Q_C = Q_{new} - Q_{old}$ ③

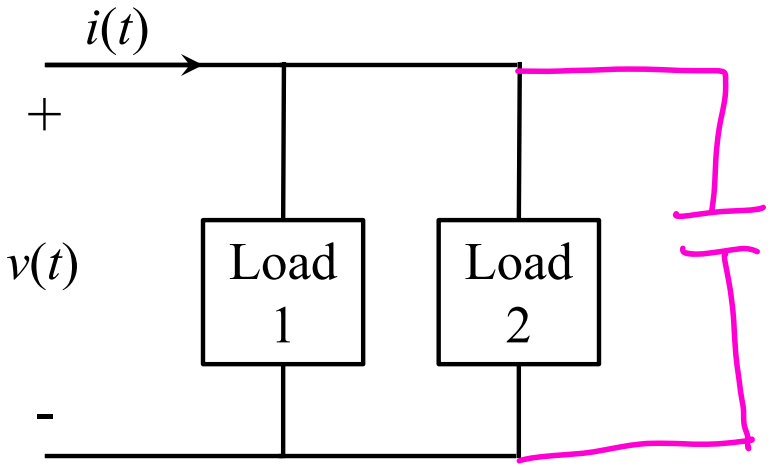
$Q_{new} = P\tan(\theta_{new}) = 1233 \tan(\theta_{new})$

$\theta_{new} = +\cos^{-1}(0.98) = 11.5^\circ$ (lagging PF)

③ $250 - 350 = -100$ VAR

② Add 100 VAR of capacitance

Example 8: PF correction



$\bar{V} = 100\angle 10^\circ \text{ V}$

$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$

$\bar{I}_2 = 5\angle 40^\circ \text{ A}$

From ex. 6

$\bar{S} = 1233 + j350 \text{ VA}$

P Qold

What should capacitor in parallel add to get PF=1?

A) 0 VAR of capacitance

B) 600 VAR of capacitance

C) 100 VAR of capacitance

D) 350 VAR of capacitance

Why 350? Because Q=0 if PF=1 (resistive load) ⇒ Need to add 350 VAR of capacitance to get Qnew=0.

2 Phase system

Single phase power:

- $p(t) = V_{rms} I_{rms} (\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i))$ → pulsating power
  ↓
  pulsating torque
  ↓
  bad for motors, noise, vibration (extra mech.) stress
  ↓
  how to fix it?
  Add another phase!

2 phase system:

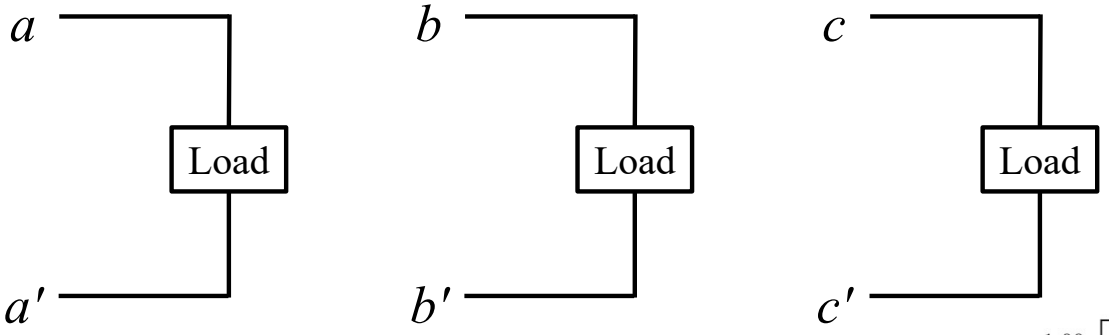
- $v_a = V_m \cos(\omega t + \theta_v)$, $i_a = I_m \cos(\omega t + \theta_i)$
- $v_b = V_m \sin(\omega t + \theta_v)$, $i_b = I_m \sin(\omega t + \theta_i)$
- $p(t) = p_a + p_b = V_m I_m \cos(\theta_v - \theta_i)$

$$P_{2\phi} = \underbrace{2 V_{rms} I_{rms} \cos(\theta)}_{P_{1\phi}}$$

constant power!

* 2x power of a single phase
* 2x wires of a single phase (4 vs 2)
* can do even better with three phase!

3 phase power



Positive Sequence:

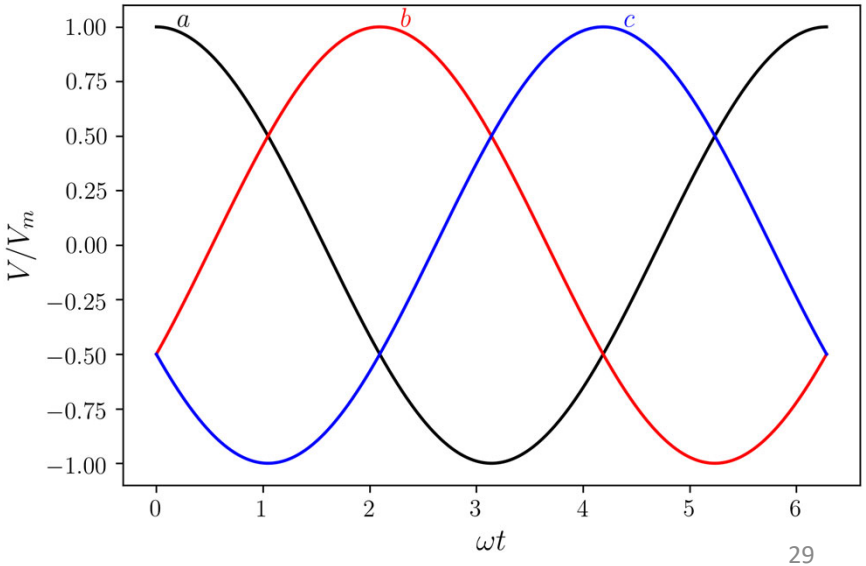
a - b - c

$$V_{aa'} = V_m \cos(\omega t)$$
$$V_{bb'} = V_m \cos(\omega t - 120^\circ)$$
$$V_{cc'} = V_m \cos(\omega t + 120^\circ)$$

Negative Sequence:

a - c - b

$$V_{aa'} = V_m \cos(\omega t)$$
$$V_{bb'} = V_m \cos(\omega t + 120^\circ)$$
$$V_{cc'} = V_m \cos(\omega t - 120^\circ)$$

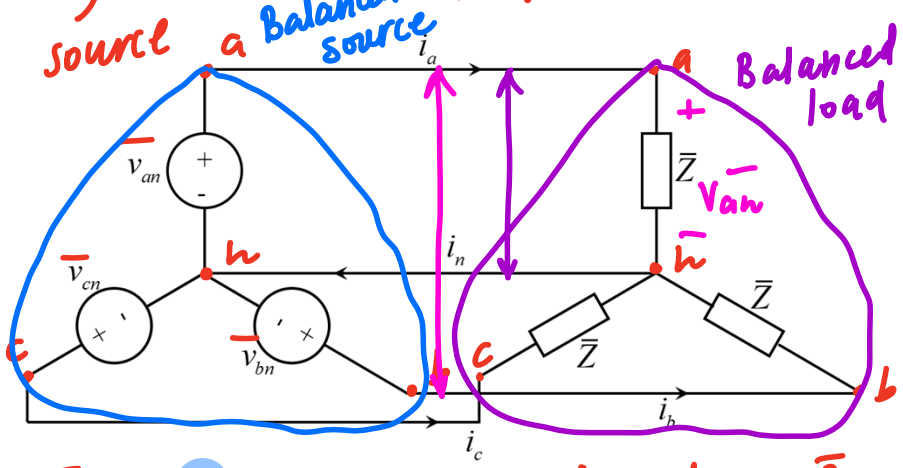


Wye-Wye Connection

source

Balanced source

load



Balanced load

$\bar{V}_{an} = V_{an} \angle 0° V$

$\bar{V}_{bn} = V_{bn} \angle -120° V$

$\bar{V}_{cn} = V_{cn} \angle 120° V$

same magnitude

impedance $\bar{Z}$ per phase is the same

$V_{ab} = V_L = \sqrt{3} V_{an} =$

$= \sqrt{3} V_{\phi,Y}$

- $|\bar{V}_{an}| = |\bar{V}_{bn}| = |\bar{V}_{cn}| = V_{\phi,Y}$ – phase voltage (phase to neutral)
- $|\bar{V}_{ab}| = |\bar{V}_{bc}| = |\bar{V}_{ca}| = V_L$ – line voltage (line – line)

Q: what is the voltage from a to b?

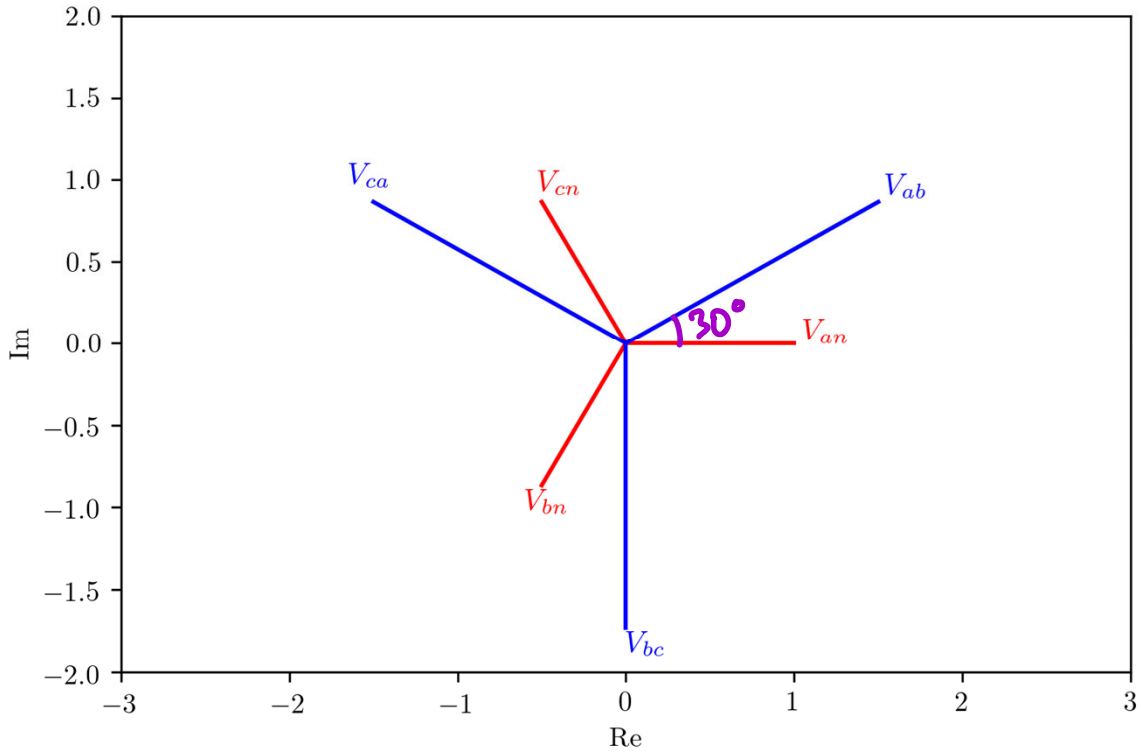
$\bar{V}_{ab} = \bar{V}_{an} - \bar{V}_{bn} = \bar{V}_{an} + (-\bar{V}_{bn})$

$\bar{V}_{ab} = V_{an} \angle 0° + V_{an} \angle \underbrace{-120° + 180°}_{60°} = \ldots$

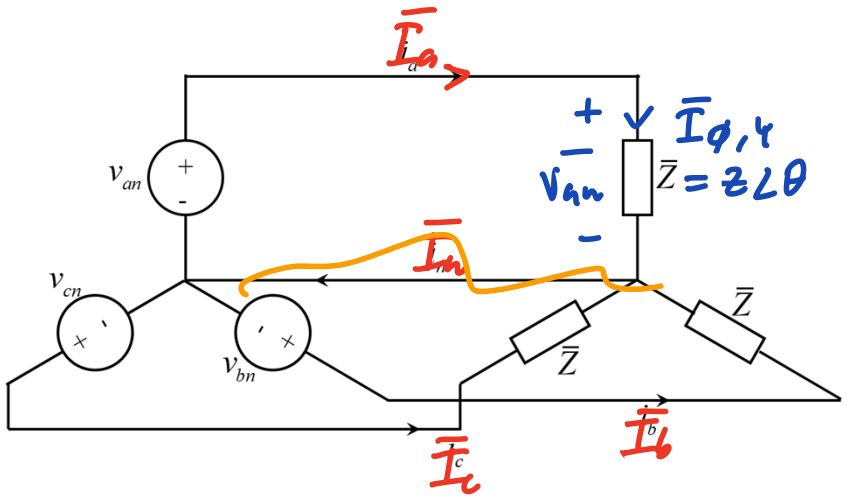
$= V_{an} \left(\frac{3}{2} + j \frac{\sqrt{3}}{2} \right)$

- $\bar{V}_{ab} = \sqrt{3} V_{an} \angle 30°$
- $\bar{V}_{bc} = \sqrt{3} V_{an} \angle -90°$
- $\bar{V}_{ca} = \sqrt{3} V_{an} \angle 150°$

Phasor Diagram



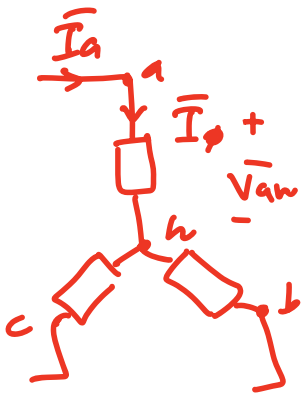
Currents in each phase/line



- Line current: current flowing in a line connecting source to a load
- Phase current: current through 1 phase at a load
- Wye-load: $\bar{I}_a = \bar{I}_L = \bar{I}_{\phi,Y}$
  - $\bar{I}_a = \dfrac{\bar{V}_{an}}{\bar{z}} = \dfrac{V_{an}\angle 0^\circ}{z\angle\theta} = I_{\phi,Y}\angle -\theta$ A
  - $\bar{I}_b = I_{\phi,Y}\angle -\theta - 120^\circ$ A
  - $\bar{I}_c = I_{\phi,Y}\angle 120^\circ - \theta$ A

Note: $\bar{I}_a + \bar{I}_b + \bar{I}_c = \bar{I}_n = 0$

Complex Power



$\bar{V}_{an} = V_{an} \angle 0° V$

$\bar{V}_{bn} = V_{bn} \angle -120° V$

$\bar{V}_{cn} = V_{cn} \angle 120° V$

$\bar{I}_a = I_L \angle -\theta$

$\bar{I}_b = I_L \angle -120° - \theta$

$\bar{I}_c = I_L \angle 120° - \theta$

- $\bar{S}_{3\phi} = \bar{S}_a + \bar{S}_b + \bar{S}_c + \bar{S}_n$ (0)
- $\bar{S}_\phi = \bar{V}_\phi \bar{I}_\phi^*$
- $\bar{S}_\phi = \bar{V}_{an} \bar{I}_a^* + \bar{V}_{bn} \bar{I}_b^* + \bar{V}_{cn} \bar{I}_c^* =$

$= V_\phi I_\phi \angle\theta + V_\phi I_\phi \angle\theta + V_\phi I_\phi \angle\theta =$

$= 3 V_\phi I_\phi \angle\theta$ ($I_\phi = I_L$, $V_\phi = V_L/\sqrt{3}$)

- $\bar{S}_{3\phi} = 3 \cdot \bar{S}_{1\phi}$

$\bar{S}_{3\phi} = \sqrt{3} V_L I_L \angle\theta$

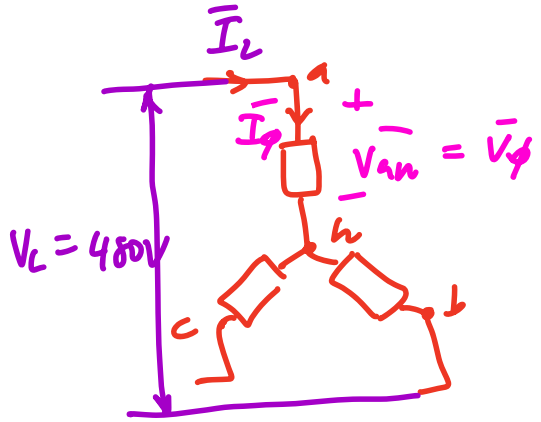
$P_{3\phi} = \sqrt{3} V_L I_L \underbrace{\cos\theta}_{PF}$

$Q_{3\phi} = \sqrt{3} V_L I_L \sin\theta$

Note: in a balanced 3ϕ load, PF of 3ϕ load is the same as PF of each phase

Example 9:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L



① $P_{3\phi} = \sqrt{3} V_L I_L \cdot PF \Rightarrow$

$$I_L = \frac{24k}{\sqrt{3} \cdot 480 \cdot 0.8} = 36.08A$$

② $P_{3\phi} = 3\,\mathrm{Re}\{\bar{V}_\phi \bar{I}_\phi^*\} =$

$= 3 V_\phi I_\phi \cos\theta = \frac{3 V_L I_L}{\sqrt{3}} \cdot \cos\theta \Rightarrow I_L = \frac{P_{3\phi}}{\sqrt{3} V_L \cdot PF} = 36.08A$

$V_\phi = \frac{V_L}{\sqrt{3}}$, $I_\phi = I_L$

Example 9-cont:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L = 36.08 A

$\theta = +\cos^{-1}(0.8) = 36.87$

$\bar{S}_{3\phi} = \sqrt{3} V_L I_L \angle\theta$

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

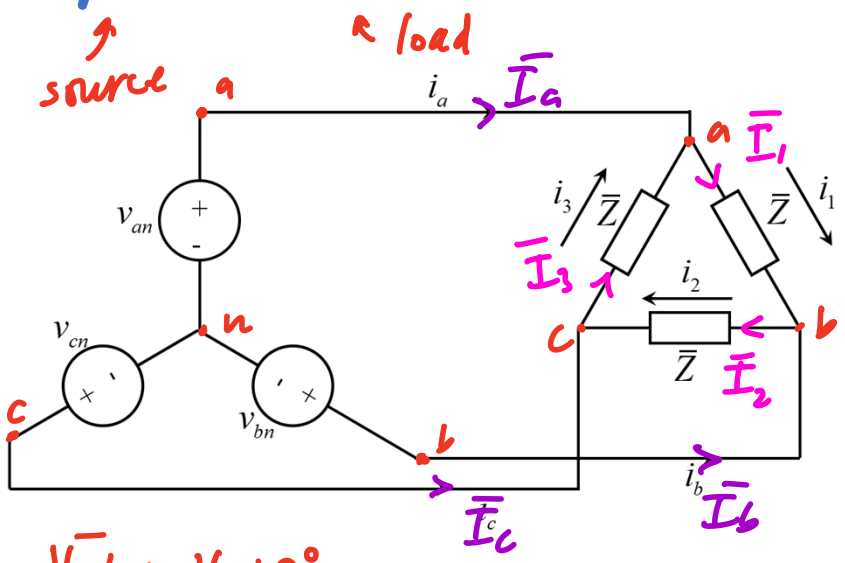
A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

Wye-Delta Connection



$\bar{V}_{ab} = V_L \angle 0°$

$\bar{V}_{bc} = V_L \angle -120°$

$\bar{V}_{ca} = V_L \angle 120°$

$V_{\phi,\Delta} = V_L = |\bar{V}_{ab}|$

$I_L = \sqrt{3}\, I_{\phi,\Delta}$

- $V_{\phi,\Delta} = V_L = |\bar{V}_{ab}|$
- $\bar{I}_1 = \dfrac{\bar{V}_{ab}}{\bar{Z}} = \dfrac{V_L \angle 0°}{Z \angle \theta} = I_{\phi,\Delta} \angle -\theta$

↑ phase current

$\bar{I}_2 = I_{\phi,\Delta} \angle -\theta - 120°$

$\bar{I}_3 = I_{\phi,\Delta} \angle -\theta + 120°$

Q: How does a line current relate to a phase current?

KCL @ (a): $\bar{I}_a = \bar{I}_1 - \bar{I}_3 = \bar{I}_1 + (-\bar{I}_3) =$

$= I_{\phi,\Delta} \angle -\theta + I_{\phi,\Delta} \angle \underbrace{120° - \theta - 180°}_{-60° - \theta}$

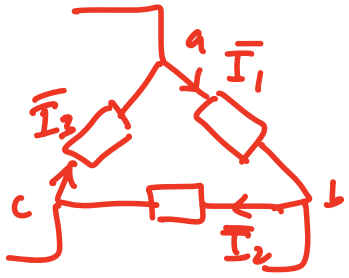
- $\bar{I}_a = \bar{I}_L = \sqrt{3}\, I_{\phi,\Delta} \angle -30° - \theta$
- $\bar{I}_b = \sqrt{3}\, I_{\phi,\Delta} \angle -150° - \theta$
- $\bar{I}_c = \sqrt{3}\, I_{\phi,\Delta} \angle -270° - \theta$

Complex Power

- $\bar{S}_{3\phi} = \bar{S}_1 + \bar{S}_2 + \bar{S}_3$
- $\bar{S}_{3\phi} = \bar{V}_{ab} \cdot \bar{I}_1^* + \bar{V}_{bc} \cdot \bar{I}_2^* +$

$+ \bar{V}_{ca} \cdot \bar{I}_3^* = 3 V_{\phi,0} \cdot I_{\phi,0} \angle\theta$

$V_{\phi,0}$ ↑ V_L ; $I_{\phi,0}$ ↑ $\frac{I_L}{\sqrt{3}}$



$\bar{V}_{ab} = V_L \angle 0°$

$\bar{V}_{bc} = V_L \angle -120°$

$\bar{V}_{ca} = V_L \angle 120°$

- $\bar{I}_1 = \frac{\bar{V}_{ab}}{\bar{Z}} = \frac{V_L \angle 0°}{Z\angle\theta} = I_{\phi,0} \angle -\theta$

↑ phase current

$\bar{I}_2 = I_{\phi,0} \angle -\theta - 120°$

$\bar{I}_3 = I_{\phi,0} \angle -\theta + 120°$

$$\bar{S}_{3\phi} = \sqrt{3} V_L I_L \angle\theta$$

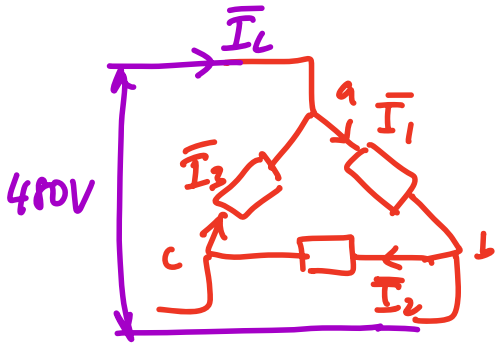
$$P_{3\phi} = \sqrt{3} V_L I_L \underbrace{\cos\theta}_{PF}$$

$$Q_{3\phi} = \sqrt{3} V_L I_L \sin\theta$$

Note: in a balanced 3ϕ load, PF of 3ϕ load is the same as PF of each phase

Example 10:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L



A) 50 A

B) 62.5 A

C) 36.08 A

D) 28.87 A

Example 10-cont:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

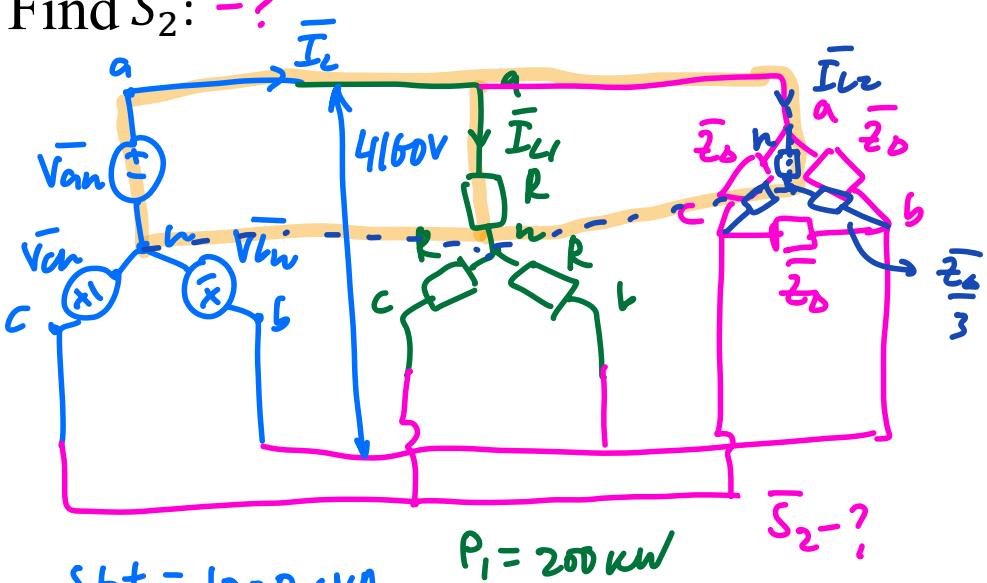
Example 11:

$S_{tot} = 1000kVa$, PF=0.8 lag, $V_L = 4160$ V

Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence $\bar{Z}_\Delta$

Find $\bar{S}_2$: -?



$S_{tot} = 1000$ kVA

PF = 0.8 lag

$\theta = +\cos^{-1}(0.8) = 36.87°$

① $\bar{S}_{tot} = \bar{S}_1 + \bar{S}_2$

$\bar{S}_2 = \bar{S}_{tot} - \bar{S}_1$ ⊖ $600 + j600$ kVA

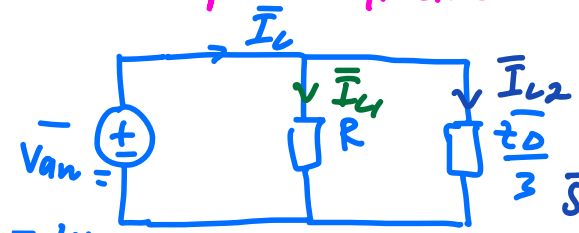
$\bar{S}_{tot}$: $1000\angle 36.87°$

$\bar{S}_1$: $P_1 + jQ_1 = 250 + j0$ kVA

P_1: 200kW

Q_1: 0 (pure resistors)

② Per-phase eq. circuit



$= \frac{4160}{\sqrt{3}}\angle 0°$ V ← reference

$P_\phi = \frac{200kW}{3}$

$S_{tot,\phi} = \frac{1000}{3}$ kVA

PF = 0.8 lag

$\bar{S}_{2,\phi} = \bar{S}_{tot,\phi} - \bar{S}_{1,\phi}$

$= 200 + j200$ kVA

$\bar{S}_{2,3\phi} = 3 \cdot \bar{S}_{2,\phi}$

$= 600 + j600$ kVA

Example 11-cont

$S_{tot} = 1000kVa$, PF=0.8 lag, $V_L = 4160$ V

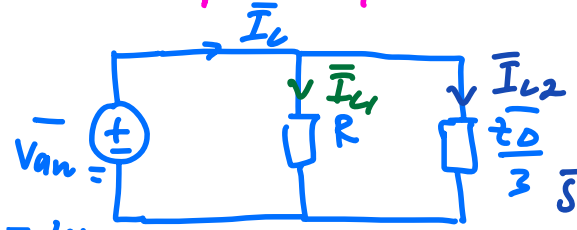
Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence

Find $\bar{S}_2$: $= 600 + j600$ kVA

Find $\bar{Z}_\Delta$: $-?$

Per-phase eq. circuit



$\bar{S}_{2,\phi} = 200 + j200 = 283\angle 45°$ kVA

① $\bar{S}_2 = \bar{V}_{an} \cdot \bar{I}_{L2}^{*} = \frac{V_{an}^2 \cdot 3}{\bar{Z}_\Delta^{*}} \Rightarrow$

$\bar{I}_{L2} = \frac{\bar{V}_{an}}{\frac{\bar{Z}_\Delta}{3}}$ ↑ $\bar{Z}_\Delta = \left(\frac{3V_{an}^2}{\bar{S}_2^{*}}\right) =$

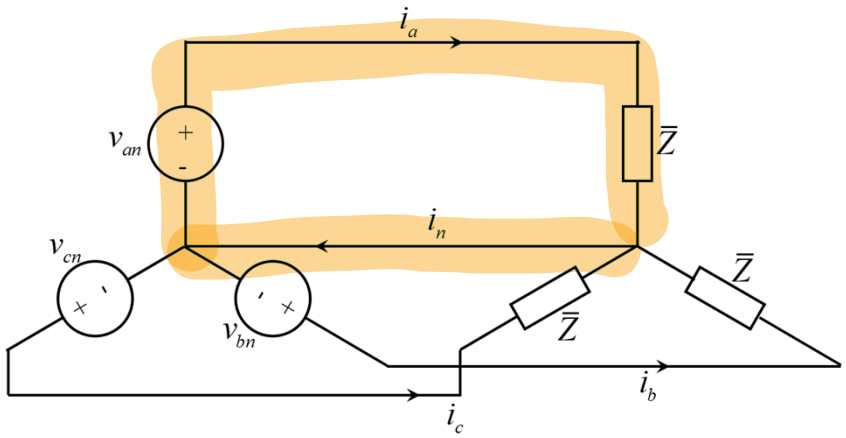
$= \frac{3 \cdot 4160^2}{283\angle -45°} =$

$= 61.15\angle 45°\ \Omega$

$S_{tot,\phi} = \frac{1000}{3}$ kVA

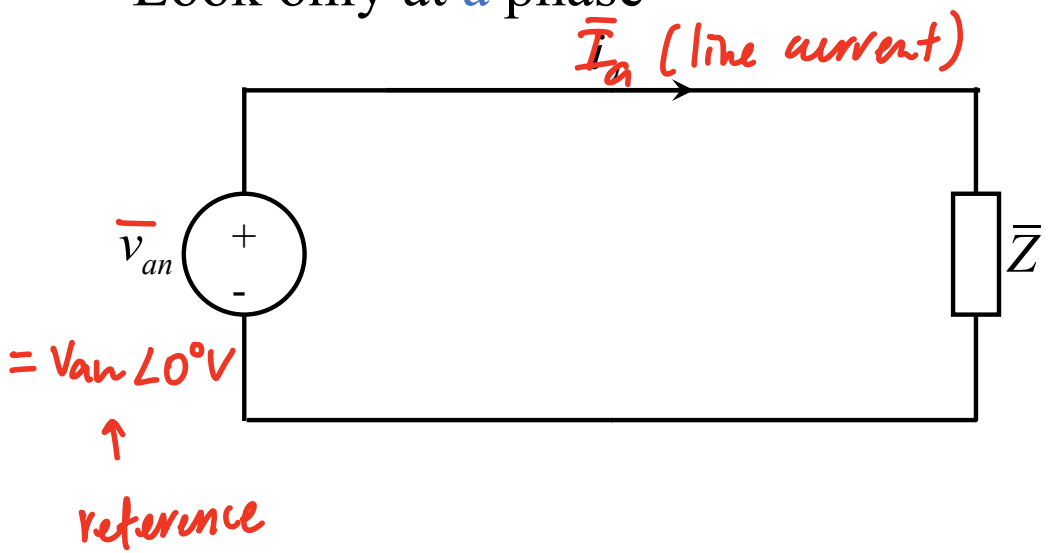
PF = 0.8 lag

Per phase equivalent circuit: Wye-Wye Connection



Per phase equivalent circuit

- Look only at a phase



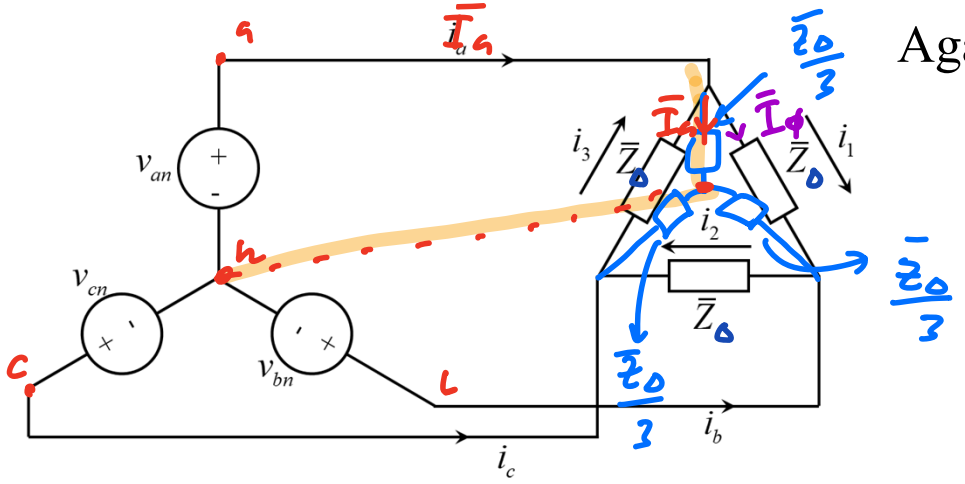
① Draw 3φ circuit

② Draw per-phase eq. circuit, solve it

③ Translate back to 3φ phase

- $\bar{S}_{3\phi} = 3\,\bar{S}_{1\phi}$

Per phase equivalent circuit: Wye-Delta Connection

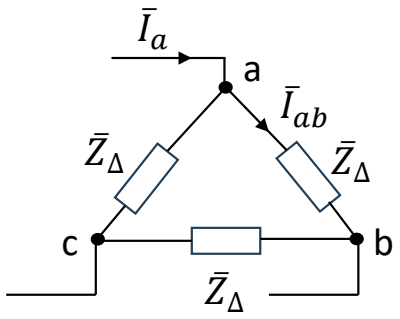


Again, only look at a phase

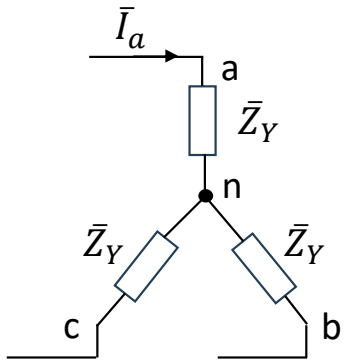
$\Delta \rightarrow Y \rightarrow 1\phi$

Delta-Wye Conversion

Delta



Wye



$$\bar{S}_{1\emptyset} = \bar{V}_{ab} \cdot \bar{I}_{ab}^{\,*} = (\bar{Z}_\Delta \cdot \bar{I}_{ab}) \cdot \bar{I}_{ab}^{\,*} = \bar{Z}_\Delta \cdot I_{ab}^{\,2} = \bar{Z}_\Delta \cdot \left(\frac{I_L}{\sqrt{3}}\right)^2$$

$\bar{Z}_\Delta \cdot \bar{I}_{ab}$ ↑

$\frac{I_a}{\sqrt{3}}$ ↑

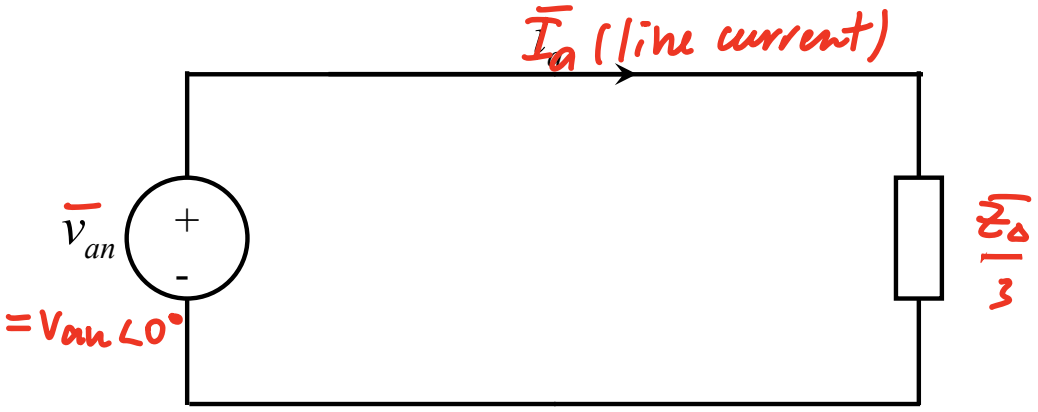
$$\bar{S}_{1\emptyset} = \bar{V}_{an} \cdot \bar{I}_a^{\,*} = (\bar{Z}_Y \cdot \bar{I}_a) \cdot \bar{I}_a^{\,*} = \bar{Z}_Y \cdot I_a^{\,2}$$

$\bar{Z}_Y \cdot \bar{I}_a$ ↑

$$\bar{S}_{1\emptyset,\Delta} = \frac{\bar{Z}_\Delta}{3} \cdot I_L^{\,2} \quad \Rightarrow \quad \bar{Z}_Y = \frac{\bar{Z}_\Delta}{3} \quad \Leftarrow \quad \bar{S}_{1\emptyset,Y} = \bar{Z}_Y \cdot I_L^{\,2}$$

To keep the same line current and complex power

Per phase equivalent circuit



① Draw 3φ circuit

② Δ → Y

③ Draw per-phase eq. circuit, solve it

④ Translate everything back to 3φ

- $\bar{S}_{3\phi} = 3\bar{S}_\phi$
- $I_{\phi,\Delta} = \frac{I_L}{\sqrt{3}}$
- $\bar{Z}_\Delta = 3\bar{Z}_Y$

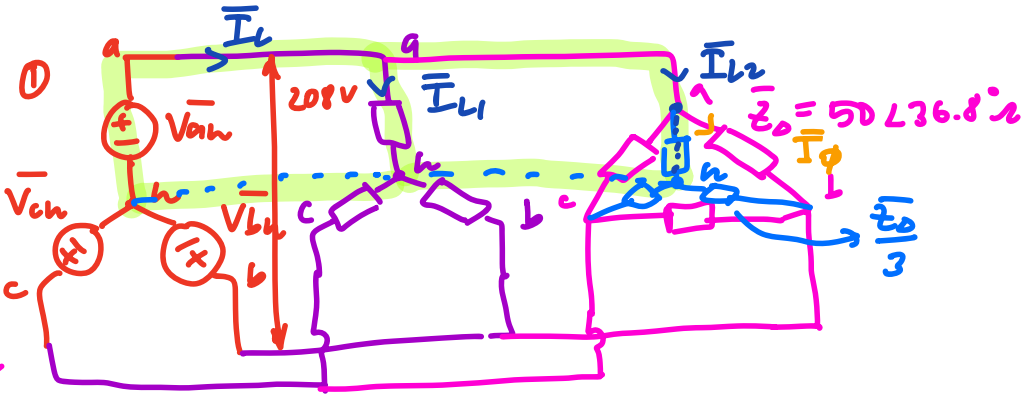
Example 12:

$V_L = 208$ V, 2 Loads in Parallel

Wye load: $P_1 = 300\,kW$, PF=0.95 lag

Delta load: $\bar{Z}_\Delta = 40 + j30\ \Omega$ $= 50\angle 36.8°\ \Omega$

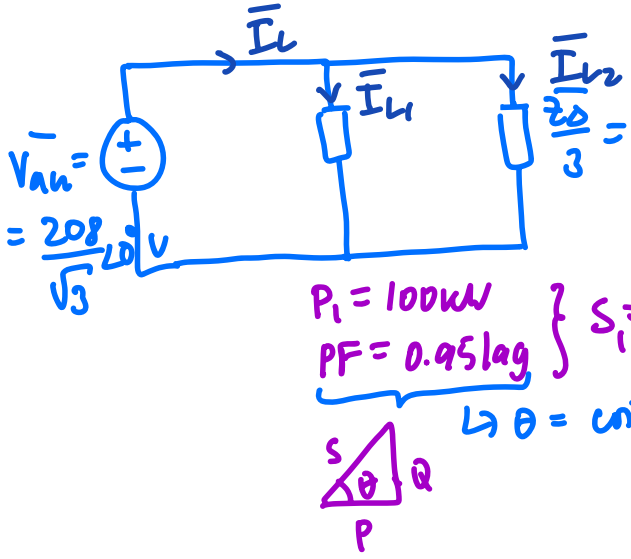
Find I_L, S_{tot}:



$P_1 = 300kW$

$PF = 0.95$ lag

② Per-phase equiv. circuit:



$P_1 = 100kW$

$PF = 0.95$ lag $\rightarrow \theta = \cos^{-1}(0.95) = 18.2°$

$S_1 = \frac{P_1}{PF} = \frac{100kW}{0.95} = 105.2\,kVA$

$\Rightarrow \bar{S}_1 = \bar{V}_{an} \cdot \bar{I}_{L_1}^* \Rightarrow \bar{I}_{L_1} = \left(\frac{\bar{S}_1}{\bar{V}_{an}}\right)^* = 876\angle -18.2°\ A$

③ $|\bar{I}_L| = |\bar{I}_{L1} + \bar{I}_{L2}|$

$\bar{I}_{L2} = \frac{\bar{V}_{an}}{\left(\frac{\bar{Z}_\Delta}{3}\right)} = 7.2\angle -36.8°\ A$

$= 882.8\angle \theta_i\ A$

+

④ $S_{tot,3\phi} = 3 \cdot S_{tot,\phi} = 3 \cdot V_{an} \cdot I_L = 318\,kVA$

Bonus: PF correction

Given: PF_old = 0.95 lag
S_tot = 318 kVA
We need to get PF_new = 0.98 lag.
Find Q_c.

$\bar{S}_{tot} = 302 + j100$ kVA

- $Q_{old} + Q_c = Q_{new}$

$Q_c = Q_{new} - Q_{old} = 61 - 100 = -39$ kVAR $\Rightarrow$ Add 39 kVAR of 3ϕ capacitance

OR $\frac{39}{3} = 13$ kVAR per phase

$Q_{new} = P\tan(\theta_{new}) = 61$ kVAR

θ_{new} from new PF